

- Q.24 Explain stroboscope?
 Q.25 Explain Load Cell?
 Q.26 Explain Ultrasonic Flowmeter?
 Q.27 Discuss bimetallic thermometer.
 Q.28 Define Pyrometry.
 Q.29 Discuss applications of pH.
 Q.30 Explain one method of torque measurement.
 Q.31 What is telemetry explain briefly?
 Q.32 Discuss X-Y recorder.
 Q.33 Explain RTD.
 Q.34 Explain electrical pressure pick-ups .
 Q.35 Discuss an electromagnetic flow meter.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain LVDT in detail with diagram. and give its application.
 Q.37 Write a short note on piezoelectric transducer, burden tube and diaphragm.
 Q.38 Write short notes on
 a) Computer monitor
 b) Dynamometer

No. of Printed Pages : 4
 Roll No.

202434

3rd Sem / Mechatronics Subject:- Electronic Instrumentation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following quantity can be directly measured by LVDT?
 a) Displacement b) Force
 c) Pressure d) Weight
- Q.2 The _____ of a strain gauge varies with applied force:
 a) Resistance b) Capacitance
 c) Inductance d) Flux
- Q.3 Which is the most common load cell used in the force sensors?
 a) Hydraulic load cells
 b) Strain-gauge based load cells
 c) Pneumatic load cells
 d) Capacitive load cells

- Q.4 What is pressure?
- The force applied over an area
 - The amount of fluid displaced by an object
 - The energy stored in a system
 - The speed of a fluid moving through a pipe
- Q.5 What principal does a strain gauge operate on
- Capacitive
 - Resistance
 - Inductive
 - Optical
- Q.6 Which of the following is type of transducer
- thermistor
 - Resistor
 - Capacitor
 - Inductor
- Q.7 Convert a temperature measurement of 250 deg C into kelvin.
- 523.2K
 - 209.7K
 - 709.7K
 - 23.2 K
- Q.8 Humidity is simply the amount of _____ held in the air.
- Water
 - water vapour
 - oxygen
 - liquid
- Q.9 A/D converter is used for _____
- converting analog to digital
 - converting digital to analog
 - converting digital to mixed signal mode
 - converting analog to mixed signal mode

- Q.10 Phase difference between two voltages at frequencies above 10Hz can be measured using

-
- CRO
 - Voltmeter
 - X-Y plotter
 - Multi meter

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define measurement.
- Q.12 What is a transducer?
- Q.13 What does LVDT stands for?
- Q.14 Define the term piezoelectricity?
- Q.15 Define absolute pressure?
- Q.16 What is Doppler Effect?
- Q.17 What is a Thermocouple?
- Q.18 What is Dew ?
- Q.19 Write the full form of DAQ.
- Q.20 Write the full form of LED.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the working of LCD. Give its advantages and disadvantages.
- Q.22 What are the various parameters considered during the selection of various transducers?
- Q.23 What is a strain gauge? Discuss in brief.